Which Distribution Fits Best? Learning Mixture Models on MNIST and Using Them to Generate New Digits

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Abstract

We compare four mixture-model families on MNIST: Gaussian Mixture Model (GMM), Laplacian Mixture Model (LMM), Bernoulli Mixture Model (BMM), and Student-t Mixture Model (SMM). Our objective is threefold: (i) fit cluster structure, (ii) quantify clustering quality, and (iii) generate new digit-like samples. The pipeline uses two preprocessing tracks: PCA-50 continuous features for GMM/LMM/SMM and binarized 784-dimensional pixels for BMM. We replaced an initially proposed VAE baseline with BMM to better match binary likelihood assumptions, and selected $K = 25$ components using BIC instead of fixing $K = 10$. Across Accuracy (Hungarian-mapped purity), ARI, and NMI, GMM is best: Accuracy = 0.7689, ARI = 0.3592, NMI = 0.6206. The main finding is that PCA-50 regularizes MNIST into a representation where Gaussian assumptions are already strong, so heavier-tailed alternatives do not outperform GMM. We also analyze convergence behavior and generated samples to compare qualitative model behavior.

1 Introduction and Motivation

Mixture models provide a principled way to represent heterogeneous data as weighted latent components. For handwritten digits, each component can capture a style mode (e.g., distinct stroke shapes for the same digit), making mixture models a natural clustering and generative framework.

This project asks: *Which distribution fits MNIST best under a common EM-based comparison?* We pursue three goals:

(1) **Fit:** learn latent clusters for handwritten digits.
(2) **Quantify:** compare models via Accuracy/Purity, ARI, and NMI.
(3) **Generate:** sample new digits from learned mixture components.

Two design updates were made relative to the proposal-stage direction. First, the proposed VAE baseline was replaced by a Bernoulli mixture model because binary-thresholded MNIST is naturally aligned with Bernoulli likelihoods and admits clear EM updates. Second, we selected $K = 25$ by Bayesian Information Criterion (BIC) analysis rather than fixing $K = 10$, allowing multiple style sub-modes per class.

We focused on four distributions (Gaussian, Laplacian, Bernoulli, Student-t) because they cover the main cases we wanted to test in one fair pipeline: standard continuous, robust continuous, heavy-tailed continuous, and binary-discrete modeling. In simple terms, GMM gives a strong baseline, Laplacian checks whether absolute-deviation style robustness helps, Student-t checks stronger heavy-tail robustness, and Bernoulli matches the binary-pixel view of MNIST. So each model had a clear purpose in the comparison, not just an extra name in the list.

We also wanted the comparison to be academically fair and practically doable. For each model, we did full EM training, convergence tracking, confusion-matrix analysis, and sample generation. If we had included many more families, we would likely have spent less effort per model and the comparison quality would have gone down. From a student project perspective, these four gave us enough diversity to answer the main question properly while keeping experiments manageable.

2 Background

2.1 Mixture Models and EM

For observations $\{x_i\}_{i=1}^N$, a K-component mixture density is

$$p(x_i) = \sum_{k=1}^{K} \pi_k p(x_i \mid \theta_k), \quad \pi_k \geq 0, \ \sum_k \pi_k = 1. \tag{1}$$

Introducing latent assignments $z_{ik} \in \{0, 1\}$, EM alternates:

$$r_{ik} \equiv p(z_{ik} = 1 \mid x_i) = \frac{\pi_k p(x_i \mid \theta_k)}{\sum_{j=1}^{K} \pi_j p(x_i \mid \theta_j)}, \tag{2}$$

$$(\pi, \theta) \leftarrow \arg\max_{\pi,\theta} \sum_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \left[\log \pi_k + \log p(x_i \mid \theta_k)\right]. \tag{3}$$

2.2 PCA Preprocessing

Principal Component Analysis (PCA) projects 784-dimensional pixels to a lower-dimensional orthogonal subspace maximizing retained variance [1]. In our implementation, $d = 50$ components provide a practical stability-speed tradeoff for continuous models.

2.3 Model Selection with BIC

BIC balances fit and complexity:

$$\text{BIC} = -2 \log \hat{L} + p \log N, \tag{4}$$

where $\hat{L}$ is maximized likelihood and p is parameter count. Lower BIC is preferred; this selected $K = 25$ in our GMM study.

Table 1: MNIST dataset summary and preprocessing used in this work.

Property	Value
Samples	70,000
Classes	10 digits (0–9)
Image size	28×28 (784 features)
Continuous track	Normalize + PCA-50
Binary track	Threshold at 0.5 (784 binary dims)
Selected components	$K = 25$ (BIC-driven)

3 Dataset and Preprocessing

We use MNIST [2] with two preprocessing tracks:

- **Track A (continuous):** normalize pixels to $[0, 1]$, then PCA to 50 dimensions for GMM/LMM/SMM.
- **Track B (binary):** threshold normalized pixels at 0.5 for BMM.

No PCA is used for Bernoulli MM because PCA creates continuous linear combinations, violating pixel-wise Bernoulli semantics.

4 Models and Methods

4.1 Gaussian Mixture Model (Diagonal Covariance)

Each component uses

$$p(x_i \mid k) = \mathcal{N}(x_i \mid \mu_k, \mathrm{diag}(\sigma_k^2)). \tag{5}$$

M-step updates are

$$N_k = \sum_i r_{ik}, \quad \pi_k = \frac{N_k}{N}, \tag{6}$$

$$\mu_k = \frac{1}{N_k} \sum_i r_{ik} x_i, \tag{7}$$

$$\sigma_k^2 = \frac{1}{N_k} \sum_i r_{ik} (x_i - \mu_k) \odot (x_i - \mu_k). \tag{8}$$

4.2 Laplacian Mixture Model

With independent Laplace dimensions,

$$p(x_i \mid k) = \prod_{d=1}^{D} \frac{1}{2b_{kd}} \exp\left(-\frac{|x_{id} - \mu_{kd}|}{b_{kd}}\right). \tag{9}$$

E-step uses the corresponding normalized responsibilities. In the M-step:

- μ_{kd} is the **weighted median** of $\{x_{id}\}_i$ under weights r_{ik}.
- $b_{kd} = \frac{1}{N_k} \sum_i r_{ik} |x_{id} - \mu_{kd}|$.

The weighted-median update is a key difference from Gaussian means.

4.3 Bernoulli Mixture Model

For binary vectors $x_i \in \{0, 1\}^D$:

$$p(x_i \mid k) = \prod_{d=1}^{D} \mu_{kd}^{x_{id}} (1 - \mu_{kd})^{1 - x_{id}}. \tag{10}$$

The M-step gives

$$\mu_{kd} = \frac{\sum_i r_{ik} x_{id}}{N_k}, \quad \pi_k = \frac{N_k}{N}. \tag{11}$$

Implementation uses log-domain computation and log-sum-exp stabilization.

4.4 Student-t Mixture Model

For robust heavy-tailed modeling, we use diagonal-covariance Student-t components:

$$p(x_i \mid k) = \mathrm{St}(x_i \mid \mu_k, \mathrm{diag}(\sigma_k^2), \nu_k). \tag{12}$$

Define

$$\delta_{ik} = \sum_d \frac{(x_{id} - \mu_{kd})^2}{\sigma_{kd}^2}, \qquad u_{ik} = \frac{\nu_k + D}{\nu_k + \delta_{ik}}. \tag{13}$$

M-step updates:

$$\mu_k = \frac{\sum_i r_{ik} u_{ik} x_i}{\sum_i r_{ik} u_{ik}}, \tag{14}$$

$$\sigma_k^2 = \frac{\sum_i r_{ik} u_{ik} (x_i - \mu_k) \odot (x_i - \mu_k)}{N_k}, \tag{15}$$

$$\pi_k = \frac{N_k}{N}. \tag{16}$$

ν_k is updated by solving a scalar fixed-point/root equation involving digamma terms (implemented with Brent root finding).

5 Evaluation

5.1 Metrics

We evaluate clustering with:

- **Accuracy/Purity** after optimal label remapping via Hungarian matching.
- **Adjusted Rand Index (ARI)**.
- **Normalized Mutual Information (NMI)**.

Let $C \in \mathbb{N}^{K \times 10}$ be the contingency matrix. Hungarian mapping maximizes

$$\max_{\Pi \in \mathcal{P}} \sum_{k=1}^{K} \sum_{j=0}^{9} C_{kj} \Pi_{kj}, \tag{17}$$

where $\mathcal{P}$ is the set of permutation/assignment matrices.

5.2 Convergence and Generative Quality

We inspect EM convergence curves (log-likelihood or lower bound trajectories), confusion matrices, and generated samples from each model as qualitative diagnostics.

6 Results

6.1 BIC-based Component Selection

We first selected the component count using BIC on the PCA-50 representation. The trend supports using $K = 25$ as a balanced point between fit quality and model complexity, and this value is used for all final comparisons.

Along with BIC, we also used a practical model-capacity check: $K = 10$ was too restrictive because each digit class has multiple writing styles (for example, crossed and uncrossed 7), while much larger K increased training time significantly across all four models

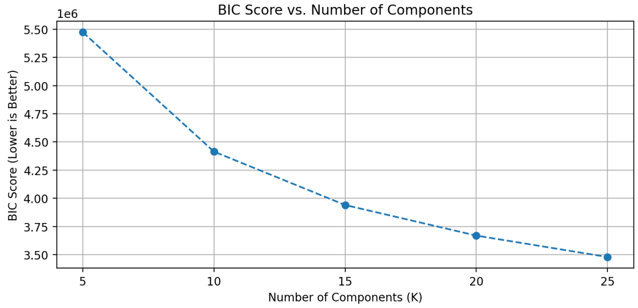


Figure 1: BIC curve used to select component count. The selected value is $K = 25$.

Table 2: Final clustering metrics (from README and comparison notebook).

Model	Accuracy	ARI	NMI
GMM	**0.7689**	**0.3592**	**0.6206**
LMM	0.7071	0.2886	0.5385
BMM	0.7595	0.3285	0.5754
SMM	0.7318	0.3133	0.5667

with only limited extra gain. In this setup, $K = 25$ worked as a good sweet spot between variation capture and computational feasibility.

6.2 Quantitative Performance

6.3 Confusion Matrices

The confusion matrices help identify which digit pairs are most frequently mixed. Most residual errors concentrate among visually similar classes, which is expected for MNIST and consistent across all four models.

6.4 Convergence Plots

Figure 3 shows that all four training procedures converge stably. The curves indicate that optimization is not the bottleneck; the final metric differences are primarily due to distributional fit.

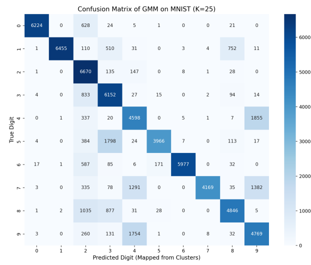
6.5 Generated Samples and Final Comparison

We also compare qualitative generation and aggregate performance visualization. The generated-sample panel, final comparison chart, and per-component panel together provide an interpretable visual summary that aligns with the metric ranking.

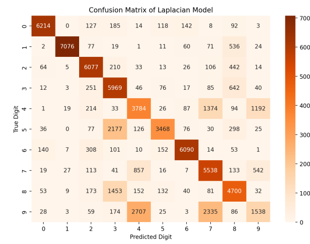
7 Discussion

GMM wins quantitatively across all three metrics. This is consistent with the geometry induced by PCA-50: dimensions are decorrelated linear combinations with smoother, near-elliptical structure, which benefits Gaussian components.

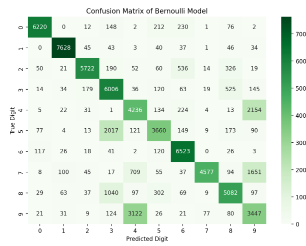
Laplacian and Student-t provide robustness in principle (median-based centers and heavy tails), but robustness does not automatically improve separation when the representation already suppresses extreme outliers. Their additional flexibility appears to be underutilized in this pipeline.



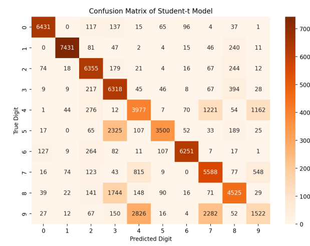
(a) GMM



(b) LMM

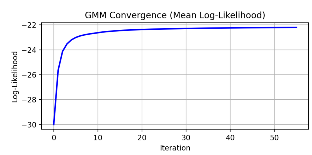


(c) BMM

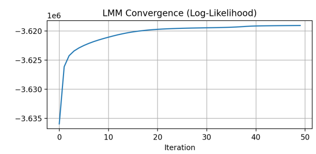


(d) SMM

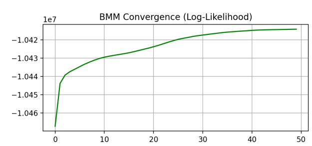
Figure 2: Confusion matrices (2×2 layout) for all four mixture models.



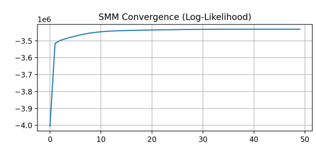
(a) GMM



(b) LMM



(c) BMM



(d) SMM

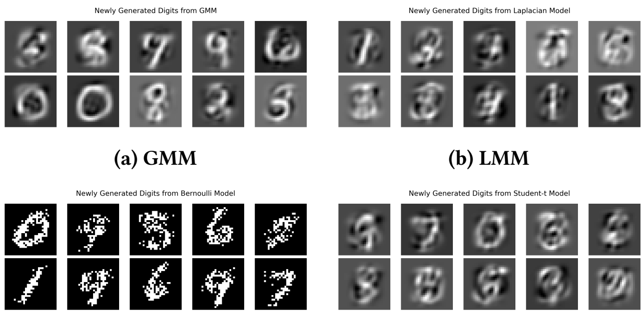
Figure 3: Convergence curves (2×2) for all models.

BMM is notably competitive on Accuracy despite strong information compression from thresholding. This indicates that coarse stroke occupancy alone still carries significant class-discriminative signal on MNIST.

Limitations. We used a single preprocessing family (PCA-50 or binary threshold), a fixed model-selection route centered on GMM-BIC, and unsupervised objectives only. More extensive hyperparameter sweeps, richer covariance structures, or alternate feature backbones may alter rankings.

8 Conclusion

We performed a controlled comparison of Gaussian, Laplacian, Bernoulli, and Student-t mixture models on MNIST for clustering and generation. With $K = 25$ chosen via BIC, GMM achieved the best overall performance (Accuracy 0.7689, ARI 0.3592, NMI 0.6206).



(a) GMM **(b) LMM**

(c) BMM **(d) SMM**

Figure 4: Generated digit samples from each model.

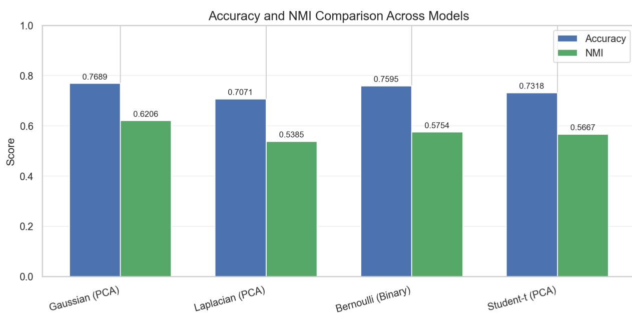


Figure 5: Final comparison chart across models.

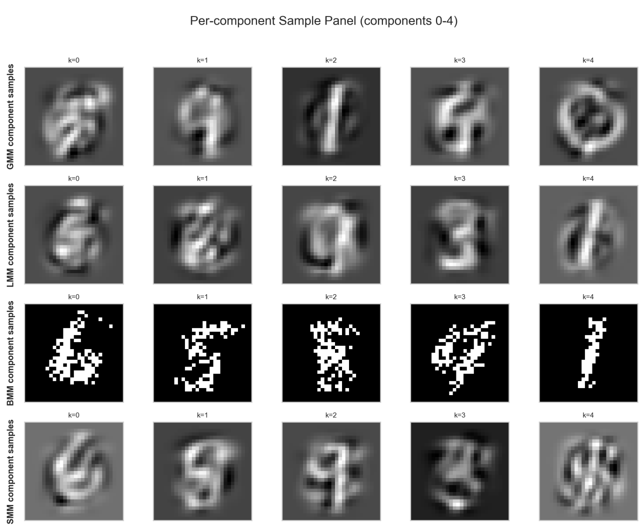


Figure 6: Per-component sample panel across all models.

The key takeaway is representation-dependent: in PCA-50 space, Gaussian assumptions are already a strong fit, making heavier-tailed alternatives less advantageous. Bernoulli MM remains a strong, interpretable baseline when binary semantics are desired.

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A Full EM Update Equations (Side-by-Side Summary)

For each model, with $N_k = \sum_i r_{ik}$:

Gaussian MM (diag):

$$r_{ik} \propto \pi_k \, \mathcal{N}(x_i \mid \mu_k, \mathrm{diag}(\sigma_k^2)), \tag{18}$$

$$\pi_k = N_k / N, \tag{19}$$

$$\mu_k = \frac{1}{N_k} \sum_i r_{ik} x_i, \tag{20}$$

$$\sigma_k^2 = \frac{1}{N_k} \sum_i r_{ik} (x_i - \mu_k) \odot (x_i - \mu_k). \tag{21}$$

Laplacian MM:

$$r_{ik} \propto \pi_k \prod_d \frac{1}{2b_{kd}} \exp\left(-\frac{|x_{id} - \mu_{kd}|}{b_{kd}}\right), \tag{22}$$

$$\pi_k = N_k / N, \tag{23}$$

$$\mu_{kd} = \mathrm{wMedian}\left(\{x_{id}\}_i; \{r_{ik}\}_i\right), \tag{24}$$

$$b_{kd} = \frac{1}{N_k} \sum_i r_{ik} |x_{id} - \mu_{kd}|. \tag{25}$$

Bernoulli MM:

$$r_{ik} \propto \pi_k \prod_d \mu_{kd}^{x_{id}} (1 - \mu_{kd})^{1 - x_{id}}, \tag{26}$$

$$\pi_k = N_k / N, \tag{27}$$

$$\mu_{kd} = \frac{1}{N_k} \sum_i r_{ik} x_{id}. \tag{28}$$

Student-t MM (diag):

$$r_{ik} \propto \pi_k \, \mathrm{St}(x_i \mid \mu_k, \mathrm{diag}(\sigma_k^2), \nu_k), \tag{29}$$

$$u_{ik} \text{ auxiliary: } u_{ik} = \frac{\nu_k + D}{\nu_k + \delta_{ik}}, \quad \delta_{ik} = \sum_d \frac{(x_{id} - \mu_{kd})^2}{\sigma_{kd}^2}, \tag{30}$$

$$\pi_k = N_k / N, \tag{31}$$

$$\mu_k = \frac{\sum_i r_{ik} u_{ik} x_i}{\sum_i r_{ik} u_{ik}}, \tag{32}$$

$$\sigma_k^2 = \frac{\sum_i r_{ik} u_{ik} (x_i - \mu_k) \odot (x_i - \mu_k)}{N_k}, \tag{33}$$

with ν_k solved each M-step via 1D root finding of the standard digamma-based equation.

References

[1] I. T. Jolliffe. 2002. *Principal Component Analysis* (2 ed.). Springer. doi:10.1007/b98835
[2] Yann LeCun, Corinna Cortes, and Christopher J. C. Burges. 1998. The MNIST Database of Handwritten Digits. (1998). https://storage.googleapis.com/cvdf-datasets/mnist/ Accessed: 2026-04-21.